On the Active Rheology of Cats: The Active Oldroyd–Felidae Constitutive Equation and a Theory of Spontaneous Egress

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Fardin's Deborah-number criterion classifies a cat as solid-like or liquid-like by comparing a feline relaxation time λ with an observation time T. It cannot, by itself, distinguish passive shape relaxation from internally powered reconfiguration, nor can it assign a probability to a cat's apparently spontaneous departure from a sink. We formulate an *Active Oldroyd–Felidae* (AOF) theory that couples an objective viscoelastic conformation tensor $\boldsymbol{C}$, a feline alignment tensor $\boldsymbol{Q}$, a conserved occupancy phase ϕ, and a stochastic metabolic activity field a. The constitutive stress $\boldsymbol{\sigma} = -p\boldsymbol{I} + 2\eta_s\boldsymbol{D} + G[f_b(\boldsymbol{C})\boldsymbol{C} - \boldsymbol{I}] - \zeta a\boldsymbol{Q} + \boldsymbol{\sigma}^K$ contains both memory and active dipolar stress. A substrate-traction closure converts internal actuation into a net wall impulse. We prove a passive energy-dissipation inequality, identify the active power that violates it, derive the finite-domain instability threshold $\zeta a > 2\Gamma(A + K_Q k_1^2)(\eta_s + G\lambda + \xi_w/k_1^2)/\xi$, and reduce sink departure to colored-noise barrier crossing with Maxwell memory. A two-state activity reduction gives a closed bi-exponential residence-time law and an exact mean exit time. Thus De answers whether a shape can relax; the AOF activity and escape numbers answer why it may rise, jump, and leave.

Dedicated to Aadhu akka, the person who introduced my love for cats.

Scope.— This is a mathematical satire in the tradition of Fardin's original note. The continuum fields below are phenomenological analogies, not claims about literal feline material composition; no animal experiment is proposed or required.

I. THE CLOSURE MISSING FROM THE DEBORAH CAT

The celebrated rheological test of Fardin [1] begins with a relaxation time λ (denoted τ there), an observation time T, and

$$\mathrm{De}_F = \frac{\lambda}{T}. \tag{1}$$

For $\mathrm{De}_F \ll 1$, shape memory decays inside the observation window and the specimen appears fluid-like; for $\mathrm{De}_F \gg 1$, it appears solid-like. This is a sound statement about *response*. It is not a dynamical closure for *agency*. Two cats may have the same λ and De_F, yet one can remain motionless while the other abruptly injects mechanical work, pushes against the basin, and exits.

Fardin explicitly observes that cats are active rheological materials and that a prescribed shear rate is therefore problematic. The issue left open is mathematical: no state variable evolves the motive power, no active stress enters the momentum equation, and no stochastic process assigns a residence-time distribution. The present theory supplies those missing objects. Its passive skeleton is Oldroyd's objective viscoelasticity [2, 3]; its powered part borrows the symmetry logic of active gels and nematics [4–7].

The conceptual separation is

$$\underbrace{\lambda}_{\text{memory decay}} \neq \underbrace{\tau_a}_{\text{activity switching}} \neq \underbrace{\tau_p}_{\text{directional persistence}}, \qquad \underbrace{\tau_p}_{\text{directional persistence}} \neq \underbrace{\mathbb{E}T_{\text{out}}}_{\text{first passage}}. \tag{2}$$

Equation (1) uses only the first. Random exit requires at least the last three.

II. KINEMATICS AND STATE VARIABLES

Let $\Omega \subset \mathbb{R}^d$, $d = 2$ or 3, contain a rigid basin $\partial\Omega_w$. The diffuse occupancy $\phi \in [0, 1]$ is one in feline material and zero in the ambient phase. The velocity is $\boldsymbol{u}$, pressure is p, and

$$\boldsymbol{D} = \tfrac{1}{2}(\nabla\boldsymbol{u} + \nabla\boldsymbol{u}^T), \qquad \boldsymbol{W} = \tfrac{1}{2}(\nabla\boldsymbol{u} - \nabla\boldsymbol{u}^T). \tag{3}$$

The symmetric positive-definite conformation tensor $\boldsymbol{C}$ stores deformation memory. The symmetric traceless tensor $\boldsymbol{Q}$ records the coarse-grained head–tail alignment relevant to an extensile or contractile active stress. A principal direction $\boldsymbol{p}$ may be selected from $\boldsymbol{Q}$ where its largest eigenvalue is simple. Finally, $a(\boldsymbol{x}, t) \geq 0$ is a dimensionless activity or motive-power field.

Mass conservation and incompressibility are written

$$\nabla \cdot \boldsymbol{u} = 0, \tag{4a}$$

$$\partial_t\phi + \nabla\cdot(\phi\boldsymbol{u}) = M_\phi\Delta\mu_\phi, \tag{4b}$$

$$\mu_\phi = \frac{\gamma}{\epsilon}W'(\phi) - \gamma\epsilon\Delta\phi, \quad W(\phi) = \tfrac{1}{4}\phi^2(1-\phi)^2. \tag{4c}$$

This Cahn–Hilliard regularization [8, 9] gives the cat–air boundary a finite thickness ϵ and interfacial energy γ without requiring a moving mesh.

The upper-convected derivative required by material-frame indifference is

$$\overset{\triangledown}{\boldsymbol{C}} = \mathrm{D}_t\boldsymbol{C} - (\nabla\boldsymbol{u})\boldsymbol{C} - \boldsymbol{C}(\nabla\boldsymbol{u})^T, \qquad \mathrm{D}_t = \partial_t + \boldsymbol{u}\cdot\nabla. \tag{5}$$

Under a superposed rigid motion $\boldsymbol{x}^* = \boldsymbol{R}(t)\boldsymbol{x} + \boldsymbol{c}(t)$, $\boldsymbol{C}^* = \boldsymbol{R}\boldsymbol{C}\boldsymbol{R}^T$ and $\overset{\triangledown}{\boldsymbol{C}}{}^* = \boldsymbol{R}\overset{\triangledown}{\boldsymbol{C}}\boldsymbol{R}^T$; a partial time derivative would fail this test.



(a) A long-lived, container-adapted state: low observed velocity does not imply passivity.



(b) An active egress event: stored and metabolically supplied work overcomes a geometric barrier.

FIG. 1. The phenomenon to be closed constitutively. In (a), the cat has conformed to a sink, consistent with $\mathrm{De}_F < 1$, yet may remain quiescent for an arbitrary interval. In (b), internally generated stress and wall traction create a finite exit impulse. Photographs from Wikimedia Commons: (a) Ken Bosma, "Cat in Sink," CC BY 2.0, source; (b) Vengolis, "Cat jump," CC BY-SA 3.0, source.

III. ACTIVE OLDROYD–FELIDAE CONSTITUTIVE LAW

A. Stress and momentum balance

The proposed total Cauchy stress is

$$\boxed{\boldsymbol{\sigma} = -p\boldsymbol{I} + 2\eta_s \boldsymbol{D} + \underbrace{G\phi[f_b(\boldsymbol{C})\boldsymbol{C} - \boldsymbol{I}]}_{\boldsymbol{\tau}^p} \underbrace{-\zeta\phi a \boldsymbol{Q}}_{-\boldsymbol{\sigma}^a} + \boldsymbol{\sigma}^K + \boldsymbol{\sigma}^Q.} \tag{6}$$

Here G is a feline elastic modulus, η_s a short-time viscosity, and

$$f_b(\boldsymbol{C}) = \frac{b-d}{b - \operatorname{tr}\boldsymbol{C}}, \qquad b > d, \tag{7}$$

is an optional FENE-P limiter; $b \to \infty$ recovers Oldroyd-B. The Korteweg stress may be taken as $\boldsymbol{\sigma}^K = -\gamma\epsilon\nabla\phi\otimes\nabla\phi$ up to an isotropic term absorbed in p. The passive orientational stress $\boldsymbol{\sigma}^Q$ is chosen consistently with the Beris–Edwards energy below [10]. The active coefficient ζ is positive for an extensile convention and negative for a contractile one.

Momentum balance, including the momentum exchange with a fixed substrate, is

$$\rho(\phi)\mathrm{D}_t\boldsymbol{u} = \nabla\cdot\boldsymbol{\sigma} + \rho(\phi)\boldsymbol{g} + \underbrace{\phi\xi_s\big(v_0 a\boldsymbol{p} - \boldsymbol{u}\big)}_{\boldsymbol{f}_{\mathrm{sub}}}. \tag{8}$$

The term $\boldsymbol{f}_{\mathrm{sub}}$ is essential. The volume integral of $\nabla\cdot\boldsymbol{\sigma}^a$ in a closed, traction-free body cannot accelerate its center of mass. A jump requires equal and opposite momentum transfer to the sink; ξ_s and v_0 are the coarse-grained substrate coupling and active slip speed.

B. Memory, alignment, and activity

The conformation evolves by

$$\overset{\triangledown}{\boldsymbol{C}} = -\frac{1}{\lambda}\big[f_b(\boldsymbol{C})\boldsymbol{C} - \boldsymbol{I}\big] + \beta_a a(\boldsymbol{Q}\boldsymbol{C} + \boldsymbol{C}\boldsymbol{Q}) + \sqrt{2D_C}\,\boldsymbol{\Xi}_C. \tag{9}$$

The first term is passive relaxation; the second is active remodeling. In deterministic calculations, positive definiteness is preserved most robustly by advancing $\boldsymbol{\Psi} = \log\boldsymbol{C}$ rather than $\boldsymbol{C}$ itself. The symmetric tensor noise $\boldsymbol{\Xi}_C$ is projected so that this constraint is not violated.

For alignment, define

$$\mathcal{F}_Q = \int_\Omega \phi\left[\frac{A}{2}\operatorname{tr}\boldsymbol{Q}^2 - \frac{B}{3}\operatorname{tr}\boldsymbol{Q}^3 + \frac{C}{4}(\operatorname{tr}\boldsymbol{Q}^2)^2 + \frac{K_Q}{2}|\nabla\boldsymbol{Q}|^2\right]\mathrm{d}V, \tag{10}$$

$$\boldsymbol{H} = -\frac{\delta\mathcal{F}_Q}{\delta\boldsymbol{Q}} + \frac{\boldsymbol{I}}{d}\operatorname{tr}\left(\frac{\delta\mathcal{F}_Q}{\delta\boldsymbol{Q}}\right). \tag{11}$$

Then

$$\mathrm{D}_t\boldsymbol{Q} - \boldsymbol{S}(\nabla\boldsymbol{u}, \boldsymbol{Q}) = \Gamma\boldsymbol{H} + \sqrt{2D_Q}\,\boldsymbol{\Xi}_Q, \tag{12}$$

with

$$\boldsymbol{S} = (\xi\boldsymbol{D} + \boldsymbol{W})(\boldsymbol{Q} + \boldsymbol{I}/d) + (\boldsymbol{Q} + \boldsymbol{I}/d)(\xi\boldsymbol{D} - \boldsymbol{W}) - 2\xi(\boldsymbol{Q} + \boldsymbol{I}/d)\operatorname{tr}(\boldsymbol{Q}\boldsymbol{D}). \tag{13}$$

The parameter ξ controls flow alignment and Γ is a rotational mobility.

The motive-power field is not prescribed; it switches stochastically:

$$\tau_a\mathrm{D}_t a = \ell_a^2\Delta a - V_a'(a) + h(\boldsymbol{x}, t) + \sqrt{2\mathcal{D}_a}\,\eta_a, \tag{14}$$

where

$$V_a(a) = \frac{\kappa_a}{4}a^2(a-a_+)^2 - \chi a. \tag{15}$$

The wells near $a = 0$ and $a = a_+$ represent quiescent and active states, respectively; χ biases wakefulness, h represents a resolvable stimulus, and η_a is space–time white noise after experimental coarse graining. A reflecting condition at $a = 0$ or the transformation $a = a_+/(1 + e^{-q})$ enforces nonnegativity. Even when $h = 0$, noise produces rare transitions and therefore unprompted activity bursts.

Equations (6)–(14), together with Eqs. (4a)–(4c), define the *Active Oldroyd–Felidae* model.

C. Wall contact and detachment

Prior to departure, the wall conditions are

$$\boldsymbol{u}\cdot\boldsymbol{n}_w = 0, \tag{16a}$$

$$\boldsymbol{P}_w(\boldsymbol{\sigma}\boldsymbol{n}_w) + \xi_w \boldsymbol{P}_w\boldsymbol{u} = \xi_w v_0 a \boldsymbol{P}_w\boldsymbol{p}, \tag{16b}$$

$$\boldsymbol{n}_w\cdot\nabla\mu_\phi = 0, \qquad \gamma\epsilon\,\boldsymbol{n}_w\cdot\nabla\phi = g_w'(\phi), \tag{16c}$$

where $\boldsymbol{P}_w = \boldsymbol{I} - \boldsymbol{n}_w\otimes\boldsymbol{n}_w$. The surface energy g_w fixes an equilibrium contact angle. Detachment is treated as a free-boundary event when the upward active work over a contact patch C exceeds adhesion plus gravity:

$$\int_{t_0}^{t_1}\int_C (\boldsymbol{\sigma}\boldsymbol{n}_w)\cdot\boldsymbol{u}\,\mathrm{d}S\mathrm{d}t > E_{\mathrm{adh}} + Mg\Delta z. \tag{17}$$

IV. A THEOREM SEPARATING RELAXATION FROM ACTIVITY

For $f_b = 1$, introduce the free energy

$$\mathcal{E} = \int_\Omega \Big[\frac{\rho}{2}|\boldsymbol{u}|^2 + \frac{G\phi}{2}\big(\mathrm{tr}\,\boldsymbol{C} - \log\det\boldsymbol{C} - d\big) + \frac{\gamma}{\epsilon}W(\phi) + \frac{\gamma\epsilon}{2}|\nabla\phi|^2 + \phi f_Q(\boldsymbol{Q},\nabla\boldsymbol{Q})\Big]\mathrm{d}V + \int_{\partial\Omega_w} g_w(\phi)\mathrm{d}S. \tag{18}$$

Proposition 1 (passive feline inequality). Assume no-flux boundaries, no external forcing, $\zeta = \beta_a = v_0 = D_C = D_Q = \mathcal{D}_a = 0$, and passive stresses consistent with Eq. (18). Smooth solutions with $\boldsymbol{C}\succ 0$ obey

$$\frac{\mathrm{d}\mathcal{E}}{\mathrm{d}t} = -\mathscr{D} \le 0, \tag{19}$$

where

$$\mathscr{D} = \int_\Omega \Big[2\eta_s|\boldsymbol{D}|^2 + \frac{G\phi}{2\lambda}\,\mathrm{tr}(\boldsymbol{C}+\boldsymbol{C}^{-1}-2\boldsymbol{I}) + M_\phi|\nabla\mu_\phi|^2 + \Gamma\phi|\boldsymbol{H}|^2 + \phi\xi_s|\boldsymbol{u}|^2\Big]\mathrm{d}V + \int_{\partial\Omega_w}\xi_w|\boldsymbol{P}_w\boldsymbol{u}|^2\mathrm{d}S. \tag{20}$$

Every term is nonnegative because $x + x^{-1} - 2 \ge 0$ for $x > 0$. Consequently an unforced passive solution cannot increase its mechanical-plus-configurational energy, although it may continue to flow while dissipating existing energy.

Proof sketch. Test momentum balance with $\boldsymbol{u}$, the phase equation with μ_ϕ, the $\boldsymbol{Q}$ equation with $-\boldsymbol{H}$, and Eq. (9) with $\frac{G\phi}{2}(\boldsymbol{I}-\boldsymbol{C}^{-1})$. Incompressibility removes pressure work; the polymeric, capillary, and orientational reversible couplings cancel pairwise. Integration by parts and Eqs. (16a)–(16c) leave Eq. (20). This is the continuum statement of passivity.

When activity is restored,

$$\frac{\mathrm{d}\mathcal{E}}{\mathrm{d}t} = -\mathscr{D} + \mathscr{P}_a + \mathcal{M}_t, \tag{21}$$

where $\mathcal{M}_t$ is a martingale/noise-injection term and the deterministic active power is, to leading order,

$$\mathscr{P}_a = \int_\Omega \phi\Big[\zeta a\boldsymbol{Q}:\boldsymbol{D} + \frac{G\beta_a a}{2}(\boldsymbol{I}-\boldsymbol{C}^{-1}):(\boldsymbol{Q}\boldsymbol{C}+\boldsymbol{C}\boldsymbol{Q}) + \xi_s v_0 a\boldsymbol{p}\cdot\boldsymbol{u}\Big]\mathrm{d}V + \int_{\partial\Omega_w}\xi_w v_0 a\,(\boldsymbol{P}_w\boldsymbol{p})\cdot(\boldsymbol{P}_w\boldsymbol{u})\,\mathrm{d}S. \tag{22}$$

Thus the instantaneous passivity index

$$\Pi_a(t) = \frac{\mathscr{P}_a(t)}{\mathscr{D}(t) + 0^+} \tag{23}$$

distinguishes a merely relaxing cat ($\Pi_a = 0$) from a powered one. Intervals with $\Pi_a > 1$ permit $\dot{\mathcal{E}} > 0$, which Eq. (1) alone forbids neither nor predicts.

V. NON-DIMENSIONAL CONTROL PARAMETERS

Choose length L, velocity U, time L/U, stress $\eta_0 U/L$, and $\eta_0 = \eta_s + G\lambda$. The leading groups are listed in Table I.

The new escape number is

$$\mathrm{Fe} = \frac{W_a}{\Delta U_s}, \qquad W_a \sim F_0 v_0 a_0^2\tau_p, \tag{24}$$

TABLE I. Dimensionless groups for active feline rheology.

Symbol	Definition	Interpretation
De_F	λ/T	observed solidity
Wi	$\lambda U/L$	elastic advection
Re	$\rho UL/\eta_0$	inertia/viscosity
Ac	$\zeta a_0/G$	active/elastic stress
Tr	$v_0 a_0/U$	active wall traction
Pe_a	$v_0 a_0 L/D_T$	persistent transport
Sp	$\tau_a U/L$	activity persistence
El	Wi/Re	elasticity/inertia
Fe	$W_a/\Delta U_s$	active work/barrier

where $\Delta U_s = E_{\rm adh} + Mg\Delta z$ is the sink barrier. The pair $(\mathrm{De}_F, \mathrm{Fe})$ resolves the logical degeneracy:

$$\mathrm{De}_F \ll 1,\ \mathrm{Fe} \ll 1 : \text{shape-adapted and resident},$$
$$\mathrm{De}_F \ll 1,\ \mathrm{Fe} \gtrsim 1 : \text{shape-adapted but able to exit},$$
$$\mathrm{De}_F \gg 1,\ \mathrm{Fe} \gtrsim 1 : \text{elastic launch or jump},$$
$$\mathrm{De}_F \gg 1,\ \mathrm{Fe} \ll 1 : \text{solid-like quiescence}.$$

No horizontal or vertical slice of this state diagram is recoverable from De_F alone.

VI. SPONTANEOUS-FLOW INSTABILITY IN A BASIN

Consider a homogeneous quiescent state $(\boldsymbol{u}, \boldsymbol{C}, \boldsymbol{Q}) = (0, \boldsymbol{I}, 0)$ with constant activity a_0. For a transverse Fourier perturbation $\{u, c, q\}e^{st+ikx}$, where $u = \delta u_y$, $c = \delta C_{xy}$, and $q = \delta Q_{xy}$, linearization gives

$$\begin{pmatrix} \rho s + \eta_s k^2 + \xi_w & -ikG & ik\zeta a_0 \\ -ik & s + \lambda^{-1} & -2\beta_a a_0 \\ -ik\xi/2 & 0 & s + r_k \end{pmatrix} \begin{pmatrix} u \\ c \\ q \end{pmatrix} = 0, \tag{25}$$

where $r_k = \Gamma(A + K_Q k^2)$. Hence the exact cubic dispersion relation is

$$\det \boldsymbol{M}(s, k) = 0. \tag{26}$$

For transparent algebra set $\beta_a = 0$. Eliminating c and q yields

$$\rho s + \xi_w + k^2 \left[\eta_s + \frac{G}{s + \lambda^{-1}} - \frac{\zeta a_0 \xi}{2(s + r_k)} \right] = 0. \tag{27}$$

The first two terms in brackets are a frequency-dependent passive viscosity; the last is a negative active viscosity. At a stationary onset $s = 0$,

$$\boxed{\zeta a_c(k) = \frac{2r_k}{\xi} \left(\eta_s + G\lambda + \frac{\xi_w}{k^2} \right).} \tag{28}$$

For a basin of effective span L_s, $k_1 \simeq \pi/L_s$ is the least stable admissible mode, so $a_0 > a_c(k_1)$ creates spontaneous circulation even without an imposed shear rate. The threshold is raised by polymer memory $G\lambda$ and wall friction ξ_w, and lowered by flow alignment ξ. Nonzero β_a couples activity directly into conformation and modifies the Routh–Hurwitz conditions in Appendix B.

The result answers the first half of the puzzle: a resident state is metastable while $a < a_c$, but an activity fluctuation crossing a_c makes rest linearly unstable. Large-scale active flow and defects are well-established consequences of similar stress–alignment feedbacks in active nematics [4, 7, 11].

VII. FROM ACTIVE STRESS TO A RANDOM JUMP TIME

A. Maxwell-memory escape coordinate

Let $z(t)$ be the vertical center-of-mass coordinate, with a sink-induced potential $U_s(z)$ having a local minimum z_m, barrier z_b, and height $\Delta U_s = U_s(z_b) - U_s(z_m)$. Projection of the continuum dynamics onto its slow exit mode produces the generalized Langevin system

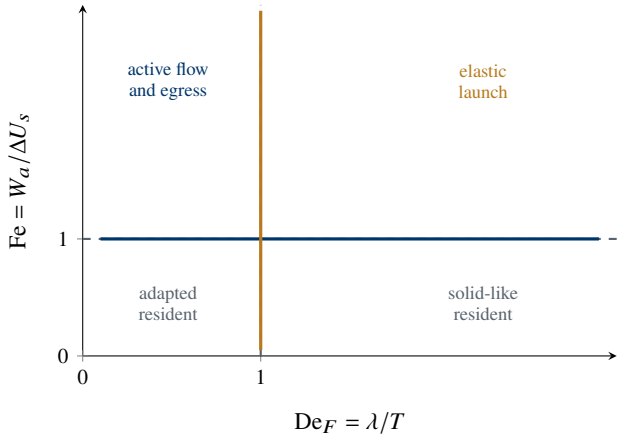


FIG. 2. Two-parameter taxonomy. The Fardin criterion supplies only the vertical division $\mathrm{De}_F = 1$; powered escape introduces the independent horizontal division $\mathrm{Fe} = 1$. Boundaries are crossovers rather than universal constants.

$$M\ddot{z} = -U_s'(z) - \gamma_s \dot{z} - r + F_0 a(t) p_z(t) + \sqrt{2\gamma_s k_B T}\, \dot{W}_s, \tag{29a}$$

$$\dot{r} = -\frac{r}{\lambda} + \frac{\gamma_p}{\lambda}\dot{z} + \frac{\sqrt{2\gamma_p k_B T}}{\lambda}\, \dot{W}_p. \tag{29b}$$

Eliminating r gives the causal kernel

$$K(t) = 2\gamma_s \delta(t) + \frac{\gamma_p}{\lambda} e^{-t/\lambda}\Theta(t), \qquad \widetilde{K}(s) = \gamma_s + \frac{\gamma_p}{1 + s\lambda}. \tag{30}$$

Thus the same λ that controls shape relaxation also filters the launch dynamics, but it does not determine when the active force $F_0 a p_z$ switches on.

Near z_m and z_b, write $U_s''(z_m) = M\omega_m^2$ and $U_s''(z_b) = -M\omega_b^2$. In the rare-escape limit, the Grote–Hynes extension of Kramers theory [12–14] gives

$$k_{\rm out}(a) \simeq \frac{\omega_m}{2\pi} \frac{\nu_b}{\omega_b} \exp\left[-\frac{\Delta U_s}{k_B T_{\rm eff}(a)} \right], \tag{31}$$

where the positive reactive frequency solves

$$M\nu_b^2 + \nu_b \left(\gamma_s + \frac{\gamma_p}{1 + \nu_b \lambda} \right) - M\omega_b^2 = 0. \tag{32}$$

If p_z has correlation time τ_p and $\langle F_0 a p_z(t) F_0 a p_z(0) \rangle = (F_0 a)^2 e^{-|t|/\tau_p}/d$, a controlled small-persistence closure motivates

$$k_B T_{\rm eff}(a; \omega_b) \approx k_B T + \frac{(F_0 a)^2 \tau_p}{d\, \gamma_{\rm eff}(\omega_b)(1 + \omega_b \tau_p)}, \qquad \gamma_{\rm eff}(\omega) = \gamma_s + \frac{\gamma_p}{1 + \omega\lambda}. \tag{33}$$

This frequency-dependent $T_{\rm eff}$ is a barrier-crossing approximation, not an equilibrium temperature: active colored noise

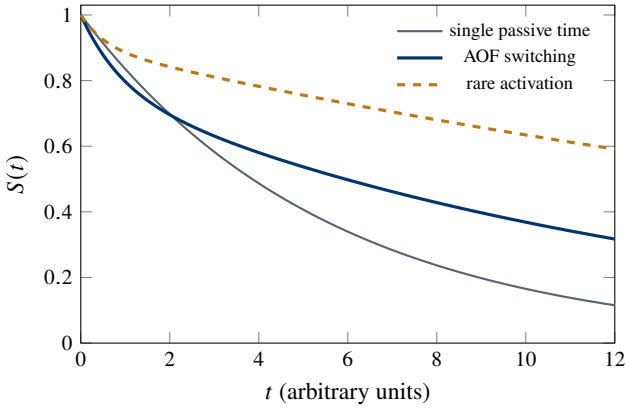


FIG. 3. Representative survival laws. A stochastic active state produces bi-exponential residence times and long tails. Curves illustrate Eq. (36); they are dimensionless predictions, not fitted animal data.

violates detailed balance [15]. Conditional on an activity history,

$$\Pr(T_{\rm out} > t \mid a) = \exp\left[-\int_0^t k_{\rm out}(a(s))\mathrm{d}s\right]. \qquad (34)$$

The exit therefore looks random even when all constitutive parameters are fixed, because the hazard is itself a stochastic functional.

B. Exact two-state residence-time law

For residence-time statistics, reduce Eq. (14) to $0 \underset{k_{\rm off}}{\overset{k_{\rm on}}{\rightleftharpoons}} 1$, with escape from the active state at rate $k_e = k_{\rm out}(a_+)$. If p_0, p_1 are the unescaped probabilities,

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix}p_0\\p_1\end{pmatrix} = \begin{pmatrix}-k_{\rm on} & k_{\rm off}\\ k_{\rm on} & -(k_{\rm off}+k_e)\end{pmatrix}\begin{pmatrix}p_0\\p_1\end{pmatrix}. \qquad (35)$$

The survival probability $S(t) = p_0(t) + p_1(t)$ is a sum of two exponentials with decay rates

$$r_\pm = \frac{1}{2}\left[\Sigma \pm \sqrt{\Sigma^2 - 4k_{\rm on}k_e}\right], \qquad \Sigma = k_{\rm on} + k_{\rm off} + k_e. \quad (36)$$

For an initially quiescent cat, the exact mean residence time is

$$\boxed{\mathbb{E}_0 T_{\rm out} = \frac{k_{\rm on}+k_{\rm off}+k_e}{k_{\rm on}k_e} = \frac{1}{k_{\rm on}} + \frac{1}{k_e} + \frac{k_{\rm off}}{k_{\rm on}k_e}.} \qquad (37)$$

The three terms have an immediate interpretation: wait to activate, wait to escape while active, and pay a penalty for aborted activation. Nothing analogous can be obtained from λ/T.

VIII. OBSERVABLE PREDICTIONS AND AN ETHICAL PROTOCOL

The theory makes falsifiable distinctions among memory, activity, and escape.

1. **Stress–activity lag.** After an activity burst, the polymeric shear stress obeys $\dot\tau^p_{xy} + \tau^p_{xy}/\lambda \simeq G\dot\gamma + 2G\beta_a aq$, so its cross-spectrum has a pole at $\omega = -i/\lambda$ distinct from the activity pole $-i/\tau_a$.

2. **Finite-size onset.** The critical activity scales as $a_c \sim A + K_Q\pi^2/L_s^2$, with an additional wall-friction term $\xi_w L_s^2/\pi^2$. Smaller basins suppress the long-wave mode but can reduce the geometric exit barrier.

3. **Non-exponential occupancy.** If activation is rate-limiting, $S(t)$ has the slow scale $r_-^{-1} \simeq (k_{\rm on}k_e/\Sigma)^{-1}$, not λ.

4. **Hysteresis.** A biased double-well V_a produces different activation and deactivation thresholds under a slowly varied stimulus h.

5. **Frequency-selective launch.** Maxwell memory reduces $\gamma_{\rm eff}(\omega_b)$ at fast barriers, altering both the Grote–Hynes transmission coefficient and active-noise filtering.

A non-invasive experiment needs only voluntary occupancy, video, and force sensing. Place an empty, stable container in a familiar environment; wait for self-loading; never restrain or startle the animal. Estimate ϕ from segmentation, $\boldsymbol{u}$ from optical flow, a from an independently specified motion/EMG proxy if available, and wall traction from a force plate. A hierarchical hidden-Markov model can fit $k_{\rm on}$, $k_{\rm off}$, and k_e across repeated voluntary visits. Relaxation after small spontaneous posture changes estimates λ independently. The decisive comparison is between a one-time-scale passive survival model and Eq. (35).



FIG. 4. A voluntary container-loading configuration suitable for passive observation. Photograph: normanack, "Box full of cat," CC BY 2.0, Wikimedia Commons source.

For parameter inference, let observations y_n at times t_n satisfy

$$y_n = \mathcal{H}[\phi, \boldsymbol{u}, \boldsymbol{Q}](t_n) + \varepsilon_n, \qquad \varepsilon_n \sim \mathcal{N}(0, \boldsymbol{R}). \tag{38}$$

With parameters $\boldsymbol{\theta} = (\lambda, \zeta, \tau_a, \mathcal{D}_a, \xi_w, \Delta U_s, \ldots)$, the posterior is

$$\pi(\boldsymbol{\theta} \mid y_{1:N}) \propto \pi_0(\boldsymbol{\theta}) \int p(y_{1:N} \mid X_{0:N}, \boldsymbol{\theta}) p(\mathrm{d}X_{0:N} \mid \boldsymbol{\theta}), \tag{39}$$

where X collects the latent continuum state. Identifiability requires at least one stress/traction observable: video alone confounds ζa, $\xi_w v_0 a$, and the barrier height.

IX. DISCUSSION

The principal result is not that a new value of λ repairs Eq. (1). It does not. Passivity is a sign constraint on the energy balance, whereas relaxation time is a memory scale. The AOF stress adds a nonconservative power channel; the activity equation switches that channel; the wall law supplies the reaction impulse; and first-passage theory converts these ingredients into a random exit time.

Three caveats sharpen the model. First, $-\zeta a \boldsymbol{Q}$ is force-free in the bulk; active stress without substrate contact can stir but cannot launch the total center of mass. Second, T_{eff} in Eq. (33) is only an asymptotic rate parameter because colored active forcing is nonequilibrium. Third, the continuum closure is intentionally over-resolved for one animal. Its value is conceptual: it exposes which observables and conservation laws any simpler model must retain.

The minimal logical chain is therefore

$$a \xrightarrow{\text{noise}} a > a_c \xrightarrow{\boldsymbol{\sigma}^a, \boldsymbol{f}_{\mathrm{sub}}} \mathscr{P}_a > \mathscr{D} \xrightarrow{W_a > \Delta U_s} T_{\mathrm{out}} < \infty. \tag{40}$$

A cat can conform to a sink at small De_F, wait because a remains in its quiescent well, and then depart when a rare activity excursion makes the resting state unstable and supplies sufficient active work. That is precisely the behavior the Deborah number classifies but cannot explain.

X. CONCLUSION

We have extended feline rheology from a passive time-scale criterion to an active stochastic constitutive theory. The resulting equations are objective, reduce to Oldroyd-B/FENE-P in the passive limit, satisfy a rigorous dissipation inequality when activity is removed, and admit spontaneous-flow and escape transitions when active power is restored. The new control variables Ac, Sp, Π_a, and Fe supplement rather than replace De_F. Most importantly, the two-state reduction yields an analytic residence-time distribution and the exact mean in Eq. (37). The sink is thus neither a proof of permanent liquidity nor a paradox of sudden solidity: it is a metastable container for an active viscoelastic body with agency encoded as a stochastic stress source.

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This paper is dedicated to Aadhu akka, whose introduction of me to the charm of cats supplied the original motive power. I also acknowledge M. A. Fardin for establishing the modern field of feline rheology. No animals were handled, constrained, or perturbed in preparing this theoretical work. Photograph creators and licenses are stated in the corresponding captions.

Appendix A: Objectivity and positivity

For a time-dependent proper orthogonal tensor $\boldsymbol{R}$, define $\boldsymbol{L} = \nabla \boldsymbol{u}$ and

$$\boldsymbol{L}^* = \dot{\boldsymbol{R}}\boldsymbol{R}^T + \boldsymbol{R}\boldsymbol{L}\boldsymbol{R}^T, \qquad \boldsymbol{C}^* = \boldsymbol{R}\boldsymbol{C}\boldsymbol{R}^T. \tag{A1}$$

Direct substitution gives

$$\mathrm{D}_t^* \boldsymbol{C}^* - \boldsymbol{L}^*\boldsymbol{C}^* - \boldsymbol{C}^*\boldsymbol{L}^{*T} = \boldsymbol{R}(\mathrm{D}_t \boldsymbol{C} - \boldsymbol{L}\boldsymbol{C} - \boldsymbol{C}\boldsymbol{L}^T)\boldsymbol{R}^T, \tag{A2}$$

proving Eq. (5) is objective. For the deterministic active remodeling equation, factor $\boldsymbol{C} = \boldsymbol{F}\boldsymbol{F}^T$ and evolve

$$\mathrm{D}_t \boldsymbol{F} = (\nabla \boldsymbol{u})\boldsymbol{F} + \beta_a a \boldsymbol{Q}\boldsymbol{F} - \frac{1}{2\lambda}\left[f_b(\boldsymbol{C})\boldsymbol{I} - \boldsymbol{C}^{-1}\right]\boldsymbol{F}. \tag{A3}$$

Then $\boldsymbol{C} \succ 0$ is automatic while $\boldsymbol{F}$ remains nonsingular. Equivalently, log-conformation methods evolve $\boldsymbol{\Psi} = \log \boldsymbol{C}$.

Appendix B: Full linear stability polynomial

Let

$$\begin{aligned} a_1 &= \lambda^{-1}, \quad a_2 = r_k, \qquad m_k = \eta_s k^2 + \xi_w, \\ Z &= \zeta a_0, \quad B_a = 2\beta_a a_0. \end{aligned} \tag{B1}$$

Expansion of Eq. (25) gives

$$\begin{aligned} 0 = {} & (\rho s + m_k)(s + a_1)(s + a_2) \\ & + k^2 G(s + a_2) - \frac{k^2 Z \xi}{2}(s + a_1) - \frac{k^2 G B_a \xi}{2}. \end{aligned} \tag{B2}$$

Writing $P(s) = b_3 s^3 + b_2 s^2 + b_1 s + b_0$, one obtains

$$b_3 = \rho, \tag{B3a}$$

$$b_2 = \rho(a_1 + a_2) + m_k, \tag{B3b}$$

$$b_1 = \rho a_1 a_2 + m_k(a_1 + a_2) + k^2 G - \frac{k^2 Z \xi}{2}, \tag{B3c}$$

$$b_0 = m_k a_1 a_2 + k^2 G a_2 - \frac{k^2 Z \xi a_1}{2} - \frac{k^2 G B_a \xi}{2}. \tag{B3d}$$

For $\rho > 0$, Routh–Hurwitz stability requires

$$b_2 > 0, \qquad b_1 > 0, \qquad b_0 > 0, \qquad b_2 b_1 > b_3 b_0. \tag{B4}$$

The stationary threshold is $b_0 = 0$, or

$$Z_c = \frac{2}{k^2 \xi a_1}\left[m_k a_1 a_2 + k^2 G a_2 - \frac{k^2 G B_a \xi}{2}\right]. \tag{B5}$$

At $B_a = 0$, this reduces exactly to Eq. (28). If $b_2 b_1 = b_3 b_0$ occurs first, the instability is oscillatory, predicting pre-jump rocking rather than steady circulation.

Appendix C: Two-state survival coefficients

For $p_0(0) = 1$, $p_1(0) = 0$, the survival law may be written explicitly as

$$S(t) = A_- e^{-r_- t} + A_+ e^{-r_+ t}, \tag{C1}$$

with

$$A_- = \frac{r_+ - k_{\mathrm{on}}}{r_+ - r_-}, \qquad A_+ = \frac{k_{\mathrm{on}} - r_-}{r_+ - r_-}. \tag{C2}$$

Because $r_- r_+ = k_{\mathrm{on}} k_e$ and $r_- + r_+ = \Sigma$, integration gives Eq. (37). The variance follows from $\mathbb{E}T^2 = 2\int_0^\infty t S(t)\mathrm{d}t$:

$$\mathrm{Var}(T_{\mathrm{out}}) = 2\left(\frac{A_-}{r_-^2} + \frac{A_+}{r_+^2}\right) - \left(\frac{A_-}{r_-} + \frac{A_+}{r_+}\right)^2. \tag{C3}$$

The squared coefficient of variation need not equal one, providing a direct statistical test against a single exponential passive model.

Appendix D: Weak form for numerical discretization

For divergence-free test velocity $\boldsymbol{v}$, symmetric tensor test $\boldsymbol{X}$, traceless symmetric $\boldsymbol{Y}$, and scalar tests ψ, α, a representative semidiscrete weak form is

$$(\rho \mathrm{D}_t \boldsymbol{u}, \boldsymbol{v}) + (2\eta_s \boldsymbol{D}, \boldsymbol{D_v}) + (\boldsymbol{\tau}^p - \zeta\phi a \boldsymbol{Q}, \nabla \boldsymbol{v}) + (\phi\xi_s \boldsymbol{u}, \boldsymbol{v}) = (\rho \boldsymbol{g} + \phi\xi_s v_0 a \boldsymbol{p}, \boldsymbol{v}), \tag{D1a}$$

$$(\overset{\triangledown}{\boldsymbol{C}}, \boldsymbol{X}) + \lambda^{-1}(f_b \boldsymbol{C} - \boldsymbol{I}, \boldsymbol{X}) - \beta_a(a(\boldsymbol{Q}\boldsymbol{C} + \boldsymbol{C}\boldsymbol{Q}), \boldsymbol{X}) = 0, \tag{D1b}$$

$$(\mathrm{D}_t \boldsymbol{Q} - \boldsymbol{S}, \boldsymbol{Y}) - \Gamma(\boldsymbol{H}, \boldsymbol{Y}) = 0, \tag{D1c}$$

$$(\partial_t \phi + \nabla \cdot (\phi \boldsymbol{u}), \psi) + M_\phi(\nabla \mu_\phi, \nabla \psi) = 0, \tag{D1d}$$

$$(\tau_a \mathrm{D}_t a, \alpha) + \ell_a^2(\nabla a, \nabla \alpha) + (V_a'(a) - h, \alpha) = 0. \tag{D1e}$$

Here $(f, g) = \int_\Omega f : g \, \mathrm{d}V$ with the obvious scalar/vector variants. A convex-splitting step for W and V_a, a log-conformation update for $\boldsymbol{C}$, and a projection step for incompressibility preserve the principal structural constraints. Stochastic increments are added after spatial covariance regularization at the imaging resolution.

[1] M.-A. Fardin, Rheology Bulletin **83**, 16 (2014).
[2] J. G. Oldroyd, Proceedings of the Royal Society of London A **200**, 523 (1950).
[3] R. B. Bird, R. C. Armstrong, and O. Hassager, *Dynamics of Polymeric Liquids, Volume 1: Fluid Mechanics*, 2nd ed. (Wiley, New York, 1987).
[4] R. A. Simha and S. Ramaswamy, Physical Review Letters **89**, 058101 (2002).
[5] Y. Hatwalne, S. Ramaswamy, M. Rao, and R. A. Simha, Physical Review Letters **92**, 118101 (2004).
[6] M. C. Marchetti, J. F. Joanny, S. Ramaswamy, T. B. Liverpool, J. Prost, M. Rao, and R. A. Simha, Reviews of Modern Physics **85**, 1143 (2013).
[7] A. Doostmohammadi, J. Ignés-Mullol, J. M. Yeomans, and F. Sagués, Nature Communications **9**, 3246 (2018).
[8] J. W. Cahn and J. E. Hilliard, Journal of Chemical Physics **28**, 258 (1958).
[9] M. E. Cates and E. Tjhung, Journal of Fluid Mechanics **836**, P1 (2018).
[10] A. N. Beris and B. J. Edwards, *Thermodynamics of Flowing Systems with Internal Microstructure* (Oxford University Press, Oxford, 1994).
[11] L. Giomi, Physical Review X **5**, 031003 (2015).
[12] H. A. Kramers, Physica **7**, 284 (1940).
[13] P. Hänggi, P. Talkner, and M. Borkovec, Reviews of Modern Physics **62**, 251 (1990).
[14] R. F. Grote and J. T. Hynes, Journal of Chemical Physics **73**, 2715 (1980).
[15] E. Fodor, C. Nardini, M. E. Cates, J. Tailleur, P. Visco, and F. van Wijland, Physical Review Letters **117**, 038103 (2016).